

BERNARDO NANDANIAR SUNIA

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Data Engineer | Data Scientist | AI/ML Engineer



Informatics graduate (GPA 3.78, Cumlaude) specializing in NLP, data analytics, and applied machine learning, with production experience building chatbots, sentiment analysis pipelines, and multilingual retrieval systems. Skilled in Python, PyTorch, and transformer-based models (BERT, SBERT), with particular depth in Indonesian-language NLP such as sarcasm, complex-negation, code-mixing, and informal text.

EDUCATION

Diponegoro University, Semarang, Indonesia

Aug 2022 – Mar 2026

Bachelor of Informatics - GPA 3.78 (Cum Laude)

WORK EXPERIENCE

YC Electric – Semarang, Indonesia

Part Time - Assistant Manager, Supply Chain Assistant, and Creative Designer

Jul 2019 – Present

- Optimized inventory tracking, streamlined supplier logistics, and produced 30+ promotional materials with digital design tools, boosting brand visibility and marketing impact.

PT Wiraky Nusa Telekomunikasi – Bandung, Indonesia

Full Time – IT Analyst

Apr 2026 – Jul 2026

- Designed and fine-tuned an Aspect Sentiment Quad Extraction (ASQE) model to extract aspects, opinions, categories, and sentiment from 5,000+ self-labeled reviews, with a focus on challenging Indonesian-language cases including sarcasm, negation, code-mixing, and informal text.
- Built and deployed an NLP-based chatbot using embedding similarity matching, with a CRUD admin dashboard and automated data-retention system for efficient knowledge-base management.
- Developed a secure review analytics system combining multi-signal anomaly detection, interactive dashboards, and a property investment analytics pipeline to support data-driven decision-making.

Class II Ambarawa Correctional Facility – Ambarawa, Indonesia

Intern - Automated Information System Chatbot Developer

May 2025 – Jul 2025

- Led end-to-end chatbot development covering stakeholder interviews, requirements definition, training dataset optimization, QA, and performance monitoring.

Computer Laboratory, Diponegoro University – Semarang, Indonesia

Laboratory Teaching Assistant

Jun 2023 – Jun 2025

- Mentored 20+ students in Python and algorithmic logic while designing structured assessments to evaluate IT competencies, contributing to a 15% improvement in average lab grades.

East Semarang District – Semarang, Indonesia

Intern - Financial Reporting Application Developer

Dec 2024 – Feb 2025

- Built a full-stack financial reporting app with ReactJS, JavaScript, and SQL, from requirements gathering to UI development and staff training, improving data accuracy and efficiency operations.

KEY PROJECT

Aspect Sentiment Classification: TF-IDF vs. SBERT on Shopee Reviews

Aug 2025 – Feb 2026

Technologies: Jupyter Notebook, Scikit-learn, Sentence-Transformers, Pandas, SciPy, Sastrawi, Statsmodels

- Built a labeled dataset from 1,000+ Shopee reviews and compared TF-IDF and SBERT representations paired with Logistic Regression, with SBERT achieving 94% F1-Score.

Custom Search Engine Using VSM and LSI

Mar 2024 – May 2024

Technologies: Python, Jupyter Notebook, Streamlit, Pandas, NumPy, Sastrawi, NLTK, scikit-learn, Matplotlib

- Developed an interactive custom search engine using VSM and LSI to improve document retrieval, combining text preprocessing, TF-IDF, cosine similarity, and semantic analysis through SVD.

Advanced Information Retrieval System Using ML & Deep Learning

Aug 2024 – Dec 2024

Technologies: Jupyter Notebook, Streamlit, Pandas, NumPy, mBERT, PyTorch, Matplotlib, Plotly

- Developed a multilingual semantic search system using fine-tuned mBERT and FAISS, enabling efficient retrieval documents through dense embedding-based similarity search.

CERTIFICATES

Intellectual Property Certificate – Software (Ministry of Law, 2026) | Intellectual Property Certificate – Chatbot (Ministry of Law, 2025) | Data Classification and Summarization (IBM SkillsBuild, 2025) | PyTorch & Generative AI (Avalon AI, 2024) | SQL & Java Programming (Oracle Academy, 2024) | Cloud Computing (Alibaba Cloud, 2024) | Data Analysis Bootcamp (Diponegoro University, 2023)

SKILLS

Programming & Tools: Python, Sentence-Transformers, Scikit-learn, FAISS, SQL, Pandas, NumPy, Streamlit, Jupyter, Git, Docker

Methods: NLP, Machine Learning, Information Retrieval, Data Analysis & Visualization, Requirements Gathering, QA Testing, Modeling

Language: English, CEFR 488, Indonesian (Native)